

A Major Project on

NAMED ENTITY RECOGNITION FOR SCANNED IMAGES, AUDIO AND VIDEO

Submitted in partial fulfilment for the
degree of Bachelor of Technology in
Data Science

Submitted by
Mansi Kumbhar (2015038)
Jayshree Parate (2015050)

Under the guidance of
Prof.Aarti Dharmani



Usha Mittal Institute of Technology
S.N.D.T. Women's University
Juhu Tara Road, Santacruz(West),
Mumbai-400049
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Certificate

This is to certify that **Mansi Kumbhar, Jayshree Parate** have completed the Major Project report on the topic **NAMED ENTITY RECOGNITION FOR SCANNED IMAGES, AUDIO AND VIDEO** satisfactorily in partial fulfilment for the Bachelor's Degree in Data Science under the guidance of **Prof.Aarti Dharmani** during the year 2023-2024 as prescribed by S.N.D.T. Women's University, Mumbai.

Guide

Prof.Aarti Dharmani

Head Of Department

Mr.Rajesh Kolte

Principal
Dr. Yogesh Nerkar

Examiner 1

Examiner 2

Abstract

Identified as a critical function within natural language processing, Named Entity Recognition (NER) entails discerning and categorizing named entities from raw textual data. This project addresses the challenge of developing an effective NER system tailored specifically for scanned images. We propose a novel approach to accurately extract text from scanned images using Optical Character Recognition (OCR) techniques and subsequently apply a context-aware NER model to identify and categorize named entities. The system is designed to handle image quality issues, consider contextual information, and provide multilanguage support. We demonstrate the system's efficiency, scalability, and accuracy in extracting valuable named entity information from scanned images, opening new possibilities for applications in document analysis, information extraction, and archival processes.

Keywords: *Named Entity Recognition (NER), Scanned Images, Optical Character Recognition (OCR), Audio Processing, Video Processing, Information Extraction.*

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Chapter 1

Introduction

Named Entity Recognition (NER) has emerged as a fundamental natural language processing (NLP) technique, initially developed to parse and categorize named entities within textual data. Traditionally, NER systems excelled at identifying entities such as persons, organizations, locations, and dates, aiding in various text-based applications like information extraction, question answering, and document summarization.

However, as the digital landscape evolves and the demand for extracting insights from diverse media formats intensifies, the limitations of conventional NER methodologies have become apparent. With the proliferation of digitized historical records, manuscripts, and printed materials, the the necessity to extend NER capabilities beyond text has become evident. Scanned images, once solely the domain of visual processing techniques, now present a unique challenge for NER systems, require them to decipher handwritten text and detect printed characters to identify named entities accurately.

Moreover, the advent of multimedia content, including audio and video has ushered in a new era for NER. In the realm of audio processing, NER must navigate the intricacies of speech recognition, transcribing spoken content, and subsequently identifying relevant entities within the audio stream. Similarly, video data presents its own set of challenges, requiring NER systems to extract textual information from visual elements like subtitles, captions, or on-screen text, and associate them

with specific entity types.

The integration of NER techniques with audio and video processing not only presents novel challenges but also opens up exciting opportunities. This fusion enables the extraction of valuable information from multimedia sources, facilitating applications such as multimedia content analysis, information retrieval, and archival processes. By extending NER beyond its traditional text-based confines to encompass audio and video, we aim to enhance the ability to identify and categorize named entities across a broader spectrum of media types, thereby unlocking new avenues for knowledge discovery and enrichment.

1.1 Problem Statement

Named Entity Recognition (NER) is a critical natural language processing task that involves identifying and classifying entities such as names of people, organizations, locations, dates, and more within unstructured text data. However, when dealing with multimedia sources such as scanned images, audio, and video, the challenges become significantly more complex.

Scanned images present numerous obstacles to traditional NER methods due to factors such as poor image quality, varied fonts and handwriting styles, noise, and distortions. Similarly, audio and video data introduce additional challenges such as speech recognition errors, background noise, accents, and diverse visual contexts. Therefore, the goal of this project is to design an innovative NER solution that can accurately extract and classify named entities not only from scanned images but also from audio and video sources.

This entails leveraging optical character recognition (OCR) techniques for text extraction in scanned images, speech recognition algorithms for transcribing audio content, and visual analysis methods for extracting textual information from video sources. Moreover, the system should possess the capability to recognize and categorize the identified text fragments into relevant entity types across all media types.

1.2 Objectives

Expanding Named Entity Recognition (NER) capabilities beyond traditional text formats represents a significant advancement in the field of information extraction and data processing. Traditional NER systems are adept at identifying and categorizing entities such as people, organizations, and locations within text. However, the digital age has seen an explosion of information in diverse media formats, including scanned images, audio, and video. To stay relevant and useful, NER systems must adapt to these changes by incorporating technologies that can process and understand non-textual data. This evolution involves the integration of optical character recognition (OCR) for images, sophisticated handwriting recognition for historical documents, and advanced speech recognition for audio content. By embracing these technologies, NER systems can extend their reach and enhance their utility across a wider array of applications.

One critical area for enhancement is the development of robust techniques for recognizing both handwritten and printed text. Historical records, manuscripts, and other digitized documents often contain handwritten text that traditional OCR systems struggle to decipher. Developing methods to accurately interpret this text can unlock vast amounts of valuable information, particularly in fields like digital humanities and archival research. Similarly, improving the detection and recognition of printed characters in scanned images will facilitate more effective named entity recognition, enabling researchers and analysts to extract meaningful data from a wealth of previously inaccessible sources.

Integrating NER with speech recognition technology represents another frontier in the evolution of these systems. By accurately transcribing spoken content and identifying relevant entities within audio streams, enhanced NER systems can support a variety of applications, from information extraction to real-time question answering. This capability is particularly valuable in domains such as media monitoring, where spoken content from news broadcasts, interviews, and podcasts can be systematically analyzed for key information. By bridging the gap between audio content and text-based data analysis, NER systems can provide deeper in-

sights and more comprehensive data analysis.

Finally, the ability to extract textual information from visual elements in video data further expands the horizons of NER applications. Developing algorithms that can recognize and analyze text in video sources—such as subtitles, captions, and on-screen text—allows for a richer and more detailed understanding of multimedia content. This capability is crucial for comprehensive multimedia content analysis and information retrieval, supporting enhanced categorization and management of data across different media types. By integrating these advanced NER techniques, we can unlock new avenues for knowledge discovery and enrichment, paving the way for innovative applications in a variety of fields, including digital humanities, multimedia content analysis, and archival research.

1.3 Motivation to do this project

The drive behind this project comes from the rapidly changing digital world, where there's a growing need for advanced tools to handle different types of media. Named Entity Recognition (NER) has been a key tool in natural language processing (NLP), especially good at finding and categorizing names and other important information in text. However, as we move forward, it's clear that traditional NER systems have limitations. This project aims to overcome these limits and take advantage of the opportunities provided by modern multimedia content.

Traditional NER systems mainly focus on text, which works well for written documents but falls short when dealing with various types of data like scanned images, audio recordings, and videos. For example, digitized historical records, manuscripts, and printed materials can be challenging because NER needs to read handwritten and printed text accurately.

The rise of multimedia content, such as audio and video, brings new challenges and opportunities for NER. Audio data requires systems that can accurately transcribe speech and then identify important names and entities within that speech. Video data adds another layer of difficulty, as it requires extracting text from ele-

ments like subtitles, captions, or on-screen text. This means NER needs to work together with advanced audio and video processing techniques to accurately recognize entities in different types of media.

Expanding NER beyond just text can open up new ways to discover and use information. Integrating NER with multimedia processing can improve applications in areas like multimedia content analysis, information retrieval, and archiving. For example, accurately identifying names in audio and video can make it easier to index and retrieve multimedia content, giving better access to valuable information.

The ultimate goal of this project is to enhance our ability to find and categorize important information from a wide range of media types. This will help us better handle the complexities of the modern digital world, leading to more effective methods for extracting and discovering information. The success of this project would mark a significant step forward for NER, bridging the gap between traditional text-based approaches and the needs of a multimedia-rich environment.

Chapter 2

Literature Survey

The text outlines various advancements and applications in the fields of computer vision, natural language processing (NLP), speech recognition, sentiment analysis, and music information retrieval (MIR). Computer vision involves the interpretation of visual data by computers, including tasks like handwriting recognition (HWR) using Python and Easy OCR [2], and optical character recognition (OCR) for extracting machine-readable text from scanned images [1]. These technologies are essential for converting handwritten or printed text into digital formats, enhancing accessibility and enabling further analysis.

NLP, a subset of artificial intelligence, focuses on machines understanding and generating natural language. It has numerous applications, including video summarization, which condenses longer videos into concise summaries using techniques such as text extraction and audio analysis [3]. Named Entity Recognition (NER) is another crucial NLP task, identifying proper names and specific terms within text, which is particularly challenging in noisy or unstructured data, such as social media posts [11].

The study of polyglots and factors influencing their language acquisition abilities is also discussed [4]. It highlights a case study of a polyglot English teacher proficient in nine languages, emphasizing the role of brain lateralization, learning strategies, age, and language contact in language acquisition. The subject's proficiency in languages, including the use of English as a bridging language, is examined along with factors such as language aptitude, motivation, and cognitive

factors contributing to successful language learning.

The development and refinement of Named Entity Recognition (NER) systems are paramount in extracting valuable information from various sources, including text, audio, and video data [7]. NER systems play a crucial role in identifying and classifying entities such as names of persons, organizations, locations, and dates [8], thus enabling deeper analysis and understanding of large volumes of unstructured data [9]. With the continuous improvement of NER algorithms and techniques [11], researchers are pushing the boundaries of what is possible in information extraction and knowledge discovery [13], paving the way for advancements in fields like natural language understanding and automated content analysis [17].

Advancements in speech recognition technologies are explored, utilizing algorithms like recurrent neural networks (RNNs) and NLP sequence models to convert spoken words into text efficiently [7]. This technology has significant applications, including transcription of speeches and audio data analysis. The development of NER in audio data [7] and end-to-end systems for extracting entities from English speech [10] showcases the progress in handling spoken language for information retrieval and processing.

Sentiment analysis aims to identify emotions in text data using classification algorithms [5]. This technology is valuable for understanding public opinion, customer feedback, and social media interactions. Comparative analysis of sentiment analysis techniques on social media data highlights the effectiveness of different approaches in capturing and interpreting emotional content [5].

In the realm of handwritten document analysis, NER is also being applied to historical texts, helping to alleviate digitization errors and improve the accuracy of extracted information [14]. This involves sophisticated techniques to handle the variability and noise inherent in handwritten documents, ensuring reliable data extraction for further historical and linguistic research [13].

These advancements illustrate the integration of various technologies to enhance data processing and analysis across different domains. From improving Optical Character Recognition (OCR) and Handwriting Recognition (HWR) [1],

to advancing speech recognition and sentiment analysis [3], and facilitating Music Information Retrieval (MIR) [6], these technologies collectively contribute to the growing field of artificial intelligence and its practical applications. By leveraging these tools, researchers and practitioners can address complex challenges in interpreting and utilizing diverse forms of data.

Lastly, the text introduces music information retrieval (MIR), a field at the intersection of musicology, digital signal processing, machine learning, and information retrieval [6]. Python has emerged as a preferred language for MIR research, driven by the availability of machine learning libraries and tools for handling text and audio data. The development of Librosa, a Python package for audio and music signal processing, aims to facilitate MIR research and make core techniques accessible to a broader community of scientists and programmers [6].

Chapter 3

Named Entity Recognition (NER)

Named Entity Recognition (NER) is a foundational task in Natural Language Processing (NLP) that involves identifying and categorizing named entities within unstructured text data. Named entities are typically real-world objects that have specific names, such as people, organizations, locations, dates, and more. NER systems analyze text and label these entities, enabling machines to understand and extract meaningful information from text data. The process of NER typically involves several steps, including tokenization, part-of-speech tagging, and entity classification.

NER systems work by first breaking down a piece of text into smaller units called tokens, such as words or phrases. These tokens are then analyzed to identify their part-of-speech (POS), which provides information about the grammatical role of each word in the sentence. Finally, machine learning algorithms are applied to classify tokens into predefined categories, such as person names, organization names, or locations. These algorithms are often trained on large annotated datasets to learn patterns and associations between words and entity types.

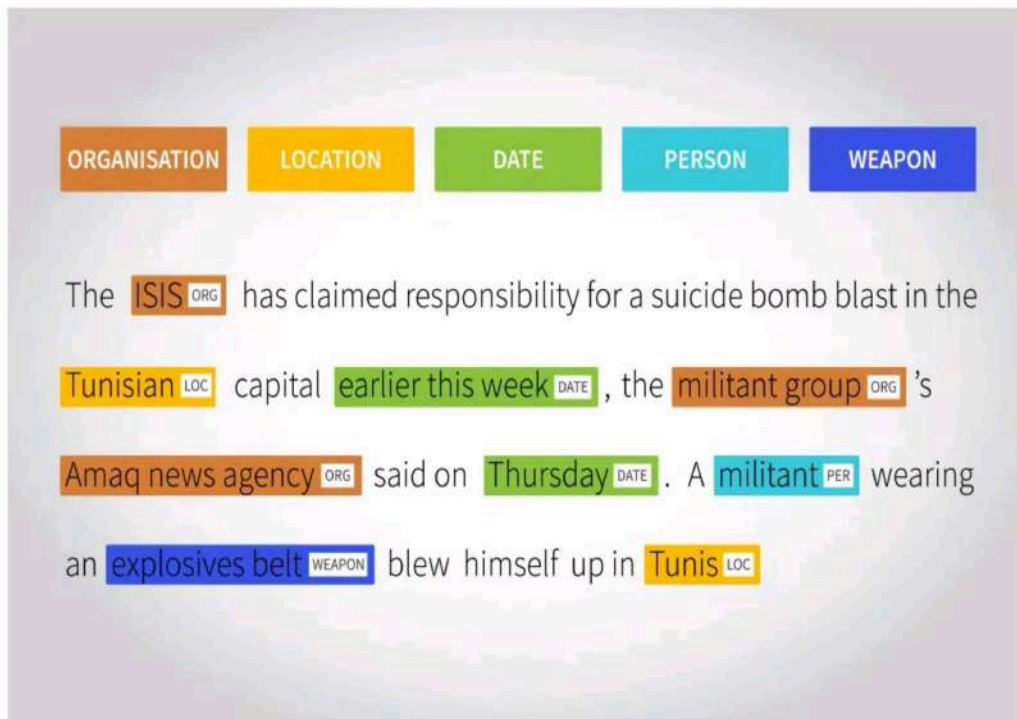


Figure 3.1: Named Entity Recognition (NER)

While NER is traditionally applied to text data, recent advancements have extended its capabilities to other forms of media, including images, audio, and video. In the context of images, NER involves detecting and categorizing objects and text within the image. Object detection techniques, such as convolutional neural networks (CNNs), are commonly used to identify objects, while optical character recognition (OCR) systems are employed to extract text from images. Once objects and text are detected, NER algorithms can classify them into relevant categories.

In the realm of audio data, NER is applied to transcribed speech to identify and categorize named entities mentioned in the audio. This process typically involves converting spoken words into text using automatic speech recognition (ASR) systems and then applying NER algorithms to the transcribed text. NER in audio data can be particularly challenging due to variations in speech patterns, accents, and background noise, requiring robust algorithms to accurately identify entities.

TWO STEPS OF THE NAMED ENTITY RECOGNITION PROCESS



Figure 3.2: Working of NER

Similarly, in video data, NER involves analyzing both visual and auditory components to identify and categorize named entities. Object detection techniques are applied to video frames to detect objects and text, while speech recognition systems transcribe spoken words into text. NER algorithms then classify these entities into relevant categories, providing a comprehensive understanding of the entities mentioned in the video.

Applications of NER in images, audio, and video span various domains, including media analysis, healthcare, security, and more. In media analysis, NER can be used to automatically annotate images, audio, and video content, enabling content recommendation systems and improving searchability. In healthcare, NER can aid in extracting relevant information from medical images, patient records, and clinical notes, facilitating clinical decision-making and research. In security and surveillance, NER can be applied to analyze video footage and audio record-

ings to identify individuals, objects, and activities of interest.

The future of NER lies in developing more robust models that can handle multilingual and cross-domain text with minimal supervision. Research is ongoing to create models that can learn from limited labeled data through techniques such as transfer learning, semi-supervised learning, and active learning. Moreover, integrating NER with other NLP tasks like sentiment analysis, relation extraction, and coreference resolution can provide a more comprehensive understanding of text, further enhancing its applications in real-world scenarios.



Figure 3.3: Working of NER

In conclusion, Named Entity Recognition is a vital NLP task that transforms unstructured text into structured information by identifying and classifying named entities. With advancements in machine learning and deep learning, NER systems have become more accurate and versatile, enabling a wide range of applications across different industries. Despite challenges like ambiguity and variation in entity mentions, ongoing research and development promise to further improve the capabilities and applications of NER in the future.

Chapter 4

Existing System

Currently, various systems exist for Named Entity Recognition (NER) across different media types such as scanned images, audio, and video. These systems leverage a combination of techniques including machine learning algorithms, deep learning models, and natural language processing (NLP) methods. For scanned images, optical character recognition (OCR) technology is often employed to convert the text within the images into machine-readable format. NER algorithms can then be applied to the extracted text to identify and classify named entities such as persons, organizations, locations, dates, and more.

In the case of audio, speech-to-text (STT) systems are utilized to transcribe spoken words into text. NER models can subsequently analyze the transcribed text to recognize named entities mentioned in the audio content. Similarly, in video processing, speech recognition techniques are applied to extract spoken words from the audio track of the video. These transcriptions, along with any on-screen text, can then be subjected to NER algorithms for entity recognition.

Here are a few existing systems:

1. Tesseract OCR:

Tesseract is an open source Optical Character Recognition (OCR) engine developed by Google, primarily designed for extracting text from scanned images with high accuracy. However, its utility extends beyond basic text extraction, as it can be integrated into more complex pipelines for tasks such as Named Entity Recognition (NER). By combining Tesseract with Natural Language Processing (NLP) techniques, users can not only digitize text from images but also analyze and categorize this text to identify and classify entities like names, dates, locations, and more. This makes Tesseract a versatile tool in the fields of document processing and data analysis, enabling more sophisticated text processing workflows.

2. Amazon Textract:

Amazon Textract is a cloud-based Optical Character Recognition (OCR) service that excels at extracting not only text but also structured data from scanned images and documents. It is capable of recognizing and pulling out entities such as names, dates, and locations from within these documents, providing a robust solution for digitizing and organizing information. Additionally, Amazon Textract can be seamlessly integrated with other Named Entity Recognition (NER) systems to enhance the extraction and classification of named entities, further streamlining workflows that require detailed data analysis and categorization. This integration makes it a powerful tool for businesses and developers looking to automate document processing and leverage extracted data for various applications.

3. Google Cloud Vision OCR :

Google Cloud Vision offers robust OCR capabilities that can efficiently extract text from scanned images, making it a valuable tool for digitizing and processing visual data. Beyond simple text extraction, Google Cloud Vision has the potential to identify and extract named entities within the text, such as names, dates, and locations. When used in combination with Named Entity Recognition (NER) models, it enhances the accuracy and depth of entity recognition, enabling more precise and comprehensive data analysis. This integration allows for the development of sophisticated applications that require detailed text understanding and

entity extraction, making Google Cloud Vision a versatile solution for a wide range of document processing and data extraction needs.

4. SpaCy:

SpaCy is a powerful open-source Natural Language Processing (NLP) library that provides pre-trained models tailored for a variety of Named Entity Recognition (NER) tasks. While SpaCy itself does not offer OCR capabilities, it excels in processing and analyzing text once it has been extracted from scanned images. By integrating SpaCy with OCR tools like Tesseract, Amazon Textract, or Google Cloud Vision, users can first digitize text from images and then apply SpaCy's advanced NER models to identify and classify entities such as names, dates, locations, and more. This combination allows for the creation of sophisticated pipelines that convert raw image data into structured and meaningful information, facilitating advanced text analysis and data extraction workflows.

5. Deep Learning Approaches:

Deep learning models, including convolutional neural networks (CNNs) and recurrent neural networks (RNNs), have demonstrated significant efficacy in performing Optical Character Recognition (OCR) and Named Entity Recognition (NER) tasks on scanned images. These models leverage the power of deep learning to accurately detect and recognize text, and subsequently, identify and classify entities within that text. Researchers frequently develop custom models tailored to their specific applications, optimizing the performance and accuracy of both OCR and NER processes. By employing CNNs' strengths in image processing and RNNs' capabilities in sequence prediction, these custom models can handle complex document layouts and varied text formats, making them highly effective for sophisticated document processing and information extraction tasks.

6. OpenNLP:

Apache OpenNLP is an open-source library designed to facilitate various natural language processing (NLP) tasks, including Named Entity Recognition (NER). While Apache OpenNLP does not offer Optical Character Recognition (OCR) capabilities, it can be effectively integrated with OCR tools such as Tesseract,

Amazon Textract, or Google Cloud Vision to process and analyze text extracted from scanned images. By using OCR tools to digitize text from images and then employing Apache OpenNLP's robust NER capabilities, users can develop comprehensive workflows that not only convert visual data into text but also accurately identify and categorize entities within that text. This integration enables the handling of complex document processing tasks, making Apache OpenNLP a valuable component in advanced text analysis and data extraction pipelines.

7. Automatic Speech Recognition (ASR) Systems :

While Automatic Speech Recognition (ASR) systems are primarily designed to transcribe spoken language into written text, they can also serve as a crucial first step in a pipeline for Named Entity Recognition (NER) from audio. Once the spoken words are accurately transcribed by the ASR system, the resulting text can be processed using standard NER techniques to identify and classify named entities such as names, dates, locations, and organizations. This integration allows for the extraction of structured information from audio data, enabling applications in various domains such as voice assistants, transcription services, and audio data analysis, thereby enhancing the accessibility and utility of spoken information.

8. Google Speech-to-Text API :

Google provides a powerful Speech-to-Text API capable of accurately transcribing spoken audio into written text. This transcribed text serves as a foundational element for further text analysis and processing. Once the audio has been converted into text, existing Named Entity Recognition (NER) libraries, such as SpaCy or NLTK, can be employed to identify and categorize named entities within the text. By combining Google's Speech-to-Text API with these advanced NER tools, users can develop sophisticated workflows that not only convert spoken language into text but also extract valuable information from it, facilitating a wide range of applications from voice-controlled interfaces to automated transcription services and comprehensive data analysis.

9. VideoBERT:

VideoBERT is a framework introduced by Sun et al. in 2019 that extends the BERT model to comprehensively understand video content. By pre-training on large-scale video datasets, VideoBERT is designed to learn representations that encode both visual and textual information simultaneously. This dual encoding capability allows VideoBERT to handle complex tasks such as Named Entity Recognition (NER) directly on video content. By integrating visual cues with associated textual data, VideoBERT can identify and classify entities within videos, enabling advanced applications in multimedia analysis, video indexing, and automated content understanding. This innovative approach leverages the strengths of transformer models in handling sequential data to bridge the gap between visual and textual information processing.

10. Open Source Projects:

In addition to specialized frameworks like VideoBERT, various open-source projects and libraries offer valuable resources for performing video analysis tasks, including Named Entity Recognition (NER). Examples such as OpenCV, TensorFlow, and PyTorch provide essential building blocks for developing custom solutions tailored to NER in video content. OpenCV offers a comprehensive suite of computer vision algorithms and tools, enabling tasks such as object detection and tracking, which can be utilized as preprocessing steps for NER. TensorFlow and PyTorch, on the other hand, provide powerful deep learning frameworks that facilitate the development of custom neural network architectures for video analysis tasks. By leveraging these libraries, developers have the flexibility to design and implement NER models that exploit both visual and textual cues within video data, contributing to advancements in multimedia understanding and content extraction.

Chapter 5

Proposed System

Named Entity Recognition (NER) has long been a cornerstone of Natural Language Processing (NLP), primarily focused on identifying and categorizing entities like people, organizations, locations, and dates within textual data. However, with the evolving digital landscape, the need to extend NER beyond mere text has become evident. This expansion is essential to effectively parse through diverse media formats such as scanned images, handwritten manuscripts, audio, and video.

In response to these challenges, a proposed system aims to enhance NER capabilities by integrating it with audio and video processing techniques. For instance, in audio processing, the system would employ speech recognition algorithms to transcribe spoken content and subsequently identify relevant entities within the audio stream. Similarly, in video data analysis, the system would extract textual information from visual elements like subtitles or captions, associating them with specific entity types.

One of the key advantages of this proposed system is its ability to decipher handwritten text and detect printed characters within scanned images. This feature is particularly crucial for historical records and printed materials, where conventional NER methodologies may fall short.

In summary, our proposed system offers valuable tools for researchers and analysts engaged in information extraction, transcription, and analysis across different media formats. By integrating NER techniques, we highlight the significance of our approach in enhancing data understanding and knowledge extraction from diverse sources. These findings contribute to the broader field of information extraction and data analysis, demonstrating the potential of NER in advancing research and exploration across various domains.

To implement this project certain software tools and devices were required are mentioned in detail below.

5.1 Software Requirements

Language : Python

Library : EasyOCR, Spacy, SpeechRecognition, Youtube-transcript-api, Wisper

Tools : Google Colab, Visual Studio

Documentation tool : Overleaf

1. Python :

Python is a high-level, interpreted programming language known for its simplicity and readability. It emphasizes code readability and allows developers to express concepts in fewer lines of code compared to other languages. Python supports multiple programming paradigms including procedural, object-oriented, and functional programming. It has a vast standard library and a thriving ecosystem of third-party packages for various tasks. Further, there is a list of Python libraries provided:

(i) Spacy :

SpaCy is an open-source natural language processing (NLP) library and toolkit written in Python. It is designed to perform various NLP tasks such as tokenization, part-of-speech tagging, named entity recognition (NER), dependency parsing, and more. SpaCy is widely used in both academic and industry due to its efficiency, accuracy, and ease of use.

One of the key features of SpaCy is its named entity recognition (NER) capabilities. NER is the process of identifying and classifying named entities in text into predefined categories such as persons, organizations, locations, dates, and more. SpaCy provides a pre-trained NER model that can be used out of the box, and it also allows you to train your own NER model on custom data if needed.

(ii) Displacy :

Displacy is a Python library and visualization tool used for visualizing the syntactic dependencies of text data. It is primarily associated with spaCy, a popular open-source natural language processing (NLP) library in Python. Displacy allows you to create visual representations of the grammatical structure of sentences or documents, making it easier to understand the relationships between words and their syntactic roles.

2. Optical Character Recognition(OCR) :

In OCR, we used EasyOCR library. It is an open-source optical character recognition (OCR) library that specializes in recognizing text from scanned images, photos, or documents. OCR technology is used to extract text information from images, making it possible to convert printed or handwritten text into machine-readable text.

3. SpeechRecognition :

Speech recognition also known as automatic speech recognition (ASR) or voice recognition, is the technology that allows a computer to transcribe spoken words into text. It involves processing audio signals and converting them into digital data that can be analyzed and understood by the computer.

4. Youtube-transcript-api :

The youtube-transcript-api is a Python library that provides an easy-to-use interface for retrieving and parsing transcripts from YouTube videos. This library allows you to extract the text transcript of a YouTube video programmatically, which can be useful for various purposes such as content analysis, data mining, or generating subtitles.

5. Wisper :

The wisper library for Python, which is used for speech recognition and audio processing tasks. This library allows you to transcribe speech from audio files into text, enabling you to work with spoken language data in your Python applications.

6. Named Entity Recognition(NER) :

After extracting text, we apply some standard NER techniques on it. Popular NER libraries and tools like spaCy, NLTK, Stanford NER, or custom NER models can be used for this purpose. These tools can identify and categorize entities such as names of people, organizations, locations, dates, and more within the text.

7. Google Colab:

Google Colab (short for Google Colaboratory) is a cloud-based platform provided by Google that allows users to write, share, and execute Python code in a collaborative environment. It provides free access to GPU and TPU resources, enabling users to train machine learning models and run data analysis tasks without needing to set up their own computational environment. Users can write code in a web-based interface similar to Jupyter notebooks and easily share their work with others.

8. Overleaf:

Overleaf is an online platform for collaborative writing and publishing of scientific documents, particularly in LaTeX, a typesetting system commonly used for academic and technical documents. It allows multiple users to work on a document simultaneously, provides templates and tools for formatting, and integrates with various publishing services. Overleaf simplifies the process of creating and sharing professional-looking documents.

Chapter 6

Implementation

The processing of various forms of media, including images, audio, and video, plays a pivotal role in numerous applications such as document management, speech recognition, and multimedia production. This process involves a series of intricate steps, each aimed at extracting valuable information from diverse sources to enable further analysis and utilization. From Optical Character Recognition (OCR) to Automatic Speech Recognition (ASR) and Named Entity Recognition (NER), these steps encompass techniques and algorithms designed to enhance the accuracy and efficiency of information extraction. In this comprehensive workflow, we explore the multifaceted process of media processing, delving into the challenges, techniques, and applications associated with each stage. Through this exploration, we uncover the underlying mechanisms driving modern media processing systems and highlight their significance in enabling automation, decision-making, and innovation across various domains.

Step 1 :

1) Input Image: Image input refers to the utilization of visual data, encompassing scanned documents, photographs, or images of handwritten text, as the primary source of information for subsequent processes. These images offer a visual representation of text or handwritten content, capturing nuances like font style, size, and layout. Their versatility allows them to be sourced from diverse mediums, including scanned documents, photographs, or digital images, making

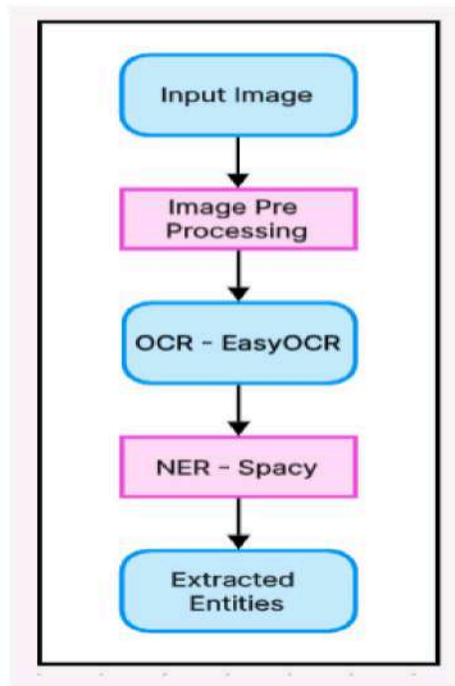


Figure 6.1: Image Flowchart

them adaptable inputs for processing systems.

However, challenges accompany image input. Variability in image quality due to factors such as resolution, lighting conditions, and distortion can significantly affect the accuracy of subsequent processing steps. Extracting text accurately from images, especially in cases of poor image quality or handwritten text, presents challenges for OCR systems. Additionally, recognition systems must contend with the diversity of languages, scripts, and writing styles present in images, necessitating robust language models and character recognition algorithms. Despite these challenges, image input remains a valuable resource for various applications, provided effective pre-processing techniques and recognition algorithms are employed.

2) Input Audio: Input audio, crucial for Automatic Speech Recognition (ASR) systems, comprises spoken language data essential for transcription. It encapsulates various characteristics and challenges. In terms of characteristics, input audio predominantly contains speech signals from one or multiple speakers, spanning conversations, lectures, phone calls, and more. Its variability presents challenges,

ranging from diverse speaker accents, speech rates, and background noise levels, to audio quality. ASR systems must exhibit robustness to handle such variations adeptly, ensuring accurate transcription. Additionally, input audio can be in multiple formats like WAV, MP3, and AAC, necessitating ASR systems' adaptability to process diverse audio file types effectively.

However, challenges persist in the realm of input audio. Background noise, environmental sounds, and overlapping speech pose significant hurdles, diminishing speech recognition accuracy. Accents and dialects further complicate matters, as ASR systems may struggle with variations outside their training data's scope, potentially leading to inaccuracies. Moreover, rapid speech or irregular pacing presents additional complexities for ASR systems to transcribe accurately.

3) Input Video: Input video, in the context of ASR (Automatic Speech Recognition), refers to the visual data accompanying audio that some systems can utilize to enhance transcription accuracy. This includes the video stream capturing facial expressions, lip movements, gestures, and other visual cues associated with speech. Video plays a crucial role in ASR by providing contextual information that aids in speech recognition. Lip movements, facial expressions, and gestures offer additional context, helping to disambiguate speech, especially in noisy environments or when dealing with unclear audio. Moreover, visual cues can assist in speaker identification and tracking, which is particularly beneficial in scenarios involving multiple speakers or poor audio quality.

However, integrating video data with audio processing pipelines presents challenges, requiring additional computational resources and specialized algorithms. Privacy concerns also arise when processing video data, especially in situations where individuals may not consent to being recorded or monitored. Furthermore, while video can improve ASR accuracy, it introduces complexities such as variations in lighting conditions, camera angles, and occlusions. Thus, while input video holds promise for enhancing ASR performance, careful consideration of these challenges and considerations is necessary for effective implementation.

Step 2 :

1) Image preprocessing is an essential initial step in image analysis tasks, aimed at enhancing the quality of raw image data before it undergoes further analysis or processing by algorithms. One fundamental aspect is noise reduction, as images often contain unwanted noise that can distort crucial features. Additionally, image enhancement techniques are utilized to adjust contrast, brightness, and sharpness, thereby improving visual quality. Normalization of intensity values is also crucial to ensure data falls within a specific range, facilitating easier comparison or analysis of images. Resizing and cropping may be necessary for standardization, while colour correction techniques such as white balancing or colour space transformations ensure an accurate representation of colours across different images.

Moreover, preprocessing involves feature extraction through methods like edge detection, corner detection, or blob detection, which extract relevant features for tasks like object recognition or image segmentation. By applying these preprocessing techniques, raw image data is refined and optimized, leading to enhanced accuracy and efficiency in subsequent image analysis algorithms.

2) The speech processing involves several steps. Initially, the system takes an audio file as input, expecting it to contain speech for transcription. Next, the input audio undergoes resampling to a standardized sample rate of 16 kHz using the librosa library. This resampling step ensures consistency in audio processing.

Subsequently, the resampled audio is processed using the Whisper model, a deep learning model specifically trained for speech recognition tasks. The Whisper model converts the resampled audio into a mel spectrogram, which is a visual representation of the audio spectrum. This mel spectrogram serves as the primary input for the speech recognition model. The Whisper model then transcribes the speech from the mel spectrogram into text format, representing the spoken content of the input audio. Following transcription, the system may perform additional processing steps such as language detection and named entity recognition, although these are not explicitly specified in the request. Finally, the system outputs the transcribed text, providing the recognized speech content extracted from the input audio file.

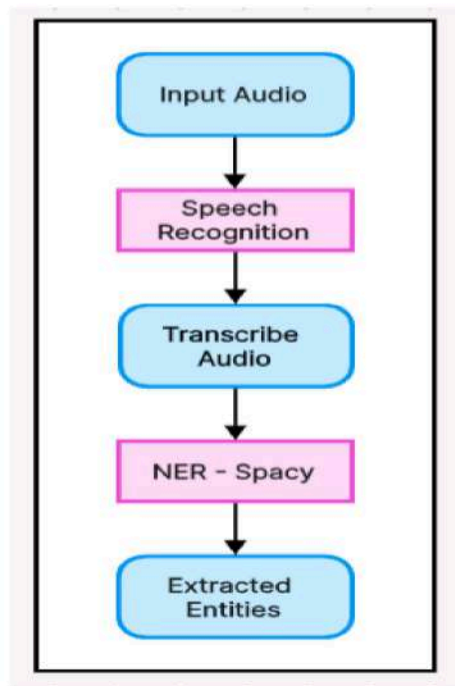


Figure 6.2: Audio Flowchart

Step 3 :

1) OCR (Optical Character Recognition) is a vital technology employed for extracting text from images. One such tool, Easy OCR, utilizes advanced algorithms to perform this task efficiently. Initially, the input image undergoes preprocessing to enhance its quality and improve recognition accuracy. Subsequently, the OCR engine employs pattern recognition techniques to analyze the image, identifying and recognizing individual characters, words, and sentences. This process involves comparing the visual patterns in the image against known character sets to decipher text accurately. Easy OCR can handle various fonts, languages, and writing styles, making them invaluable for digitizing printed documents, automating data entry tasks, and enabling text extraction from images in diverse applications such as document management, digital archives, and text analytics.

2) In the stage of speech recognition, specific characteristics of the speech audio are extracted to aid in distinguishing between different phonemes, which are the fundamental units of sound in a language. These extracted features, such as pitch

and formants, play a crucial role in enabling the system to discern nuances in speech patterns. By analyzing these features, the speech recognition system can accurately interpret spoken words and phrases, facilitating effective communication between humans and machines. This process of feature extraction serves as a foundational step in the broader field of natural language processing, enabling advancements in various applications ranging from virtual assistants to automated transcription services.

Step 4 :

1) Extract Text (from OCR): Text extraction is a critical stage in the process of scanning images via Optical Character Recognition (OCR). When text is identified within the scanned image, this stage becomes essential for extracting that text data. By extracting the text, it becomes usable and accessible for various purposes such as indexing, searching, and analysis. This process enables the conversion of scanned documents into editable and searchable formats, facilitating easier information retrieval and manipulation. Additionally, text extraction enhances the efficiency of data processing and management by digitizing content that was previously only available in non-editable formats. Through OCR-based text extraction, organizations and individuals can unlock the potential of scanned documents, making them more versatile and valuable in the digital age.

2) Transcribe Audio: In the process of speech recognition, features extracted from the speech audio are pivotal. These features serve as the foundation for a speech recognition model to determine the most probable sequence of words corresponding to the audio input. By leveraging these extracted features, the model analyzes the acoustic properties of the speech, identifying patterns and nuances that aid in accurate transcription. Through this intricate process, the model deciphers the spoken language, converting it into understandable text. This integration of feature extraction and speech recognition plays a crucial role in enabling machines to comprehend and interpret human speech effectively.

3) Extract Audio (from Video): Extracting audio from a video is a crucial step in many multimedia tasks. When a video is inputted, the process involves separating the audio stream from the video stream. This separation allows for

independent manipulation of the audio, such as editing, mixing, or applying effects. By extracting the audio, users can repurpose it for various purposes, such as creating podcasts, music tracks, or sound effects libraries. Additionally, it enables efficient storage and transmission of audio data, particularly in scenarios where only the audio content is needed, saving bandwidth and disk space. Overall, extracting audio from a video stream enhances flexibility and usability in multimedia production and distribution.

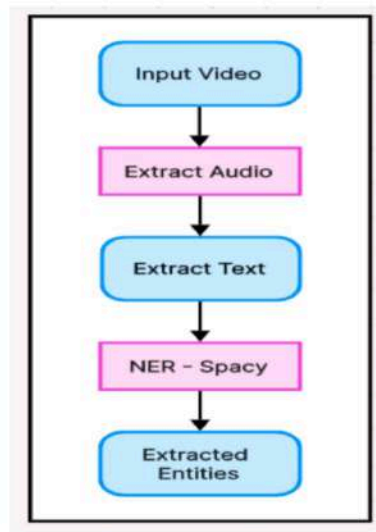


Figure 6.3: Video Flowchart

Step 5 :

Named Entity Recognition (NER) - SpaCy: Named Entity Recognition (NER) is a crucial step in speech recognition systems like SpaCy, where it identifies and categorizes named entities within the transcribed text. These entities encompass various elements such as people's names, locations, or organizations. Although not all automatic speech recognition (ASR) systems incorporate this step, NER plays a significant role in extracting pertinent information from the transcribed content, facilitating tasks like information retrieval and analysis. By accurately identifying and classifying named entities, NER enhances the overall understanding and usability of transcribed speech data.

Step 6 :

The final output of the process is a list of extracted entities from the input image. These entities can include names of people, organizations, locations, dates, numbers, and other relevant information present in the text, audio, and video. This is the final output, which can include the transcribed text, and any recognized named entities. Extracted entities provide crucial information that aids in various tasks such as information retrieval, data analysis, and content summarization. By identifying and organizing these entities, systems can efficiently process and interpret the underlying data, facilitating seamless integration into workflows and applications. This output serves as a valuable resource for understanding and utilizing the content contained within the input, contributing to enhanced automation and decision-making processes across different domains and industries.

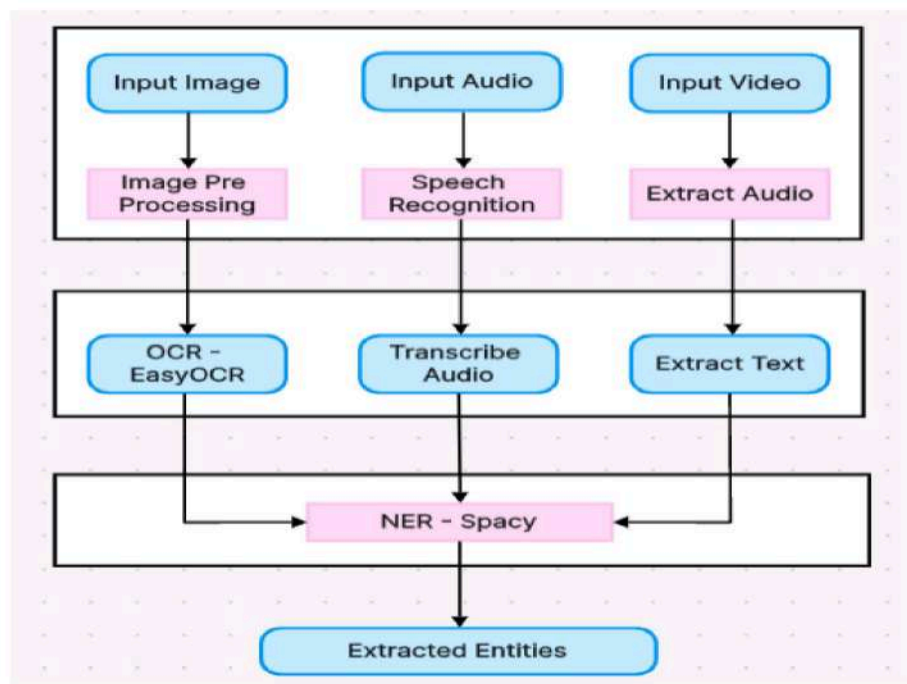


Figure 6.4: Complete Working of Project

Chapter 7

Results

```
[34] NER = spacy.load("en_core_web_sm")

[35] raw_text="The Indian Space Research Organisation or is the national space agency of India, headquartered in Bengaluru. It operates under Department of Space which is directly overseen by the Prime Minister of India while Chairman of ISRO acts as executive of DOS as well."

[36] text1=NER(raw_text)

[37] for word in text1.ents:
    print(word.text,word.label_)

The Indian Space Research Organisation ORG
India GPE
Bengaluru GPE
Department of Space ORG
India GPE
ISRO ORG
DOS ORG

[38] spacy.explain("ORG")

'Companies, agencies, institutions, etc.'
```

The Indian Space Research Organisation ORG or is the national space agency of India GPE headquartered in Bengaluru GPE It operates under Department of Space ORG which is directly overseen by the Prime Minister of India GPE while Chairman of ISRO ORG acts as executive of DOS ORG as well.

Figure 7.1: Output of Text NER

The provided Python code uses the spaCy library for named entity recognition (NER), a technique in natural language processing that identifies and categorizes entities in text, such as names of people, organizations, locations, dates, and more. The code loads a pre-trained English model (en-core-web-sm) and processes a text mentioning the Indian Space Research Organisation (ISRO). It identifies "ISRO" and "The Department of Space (DOS)" as organizations (ORG), and "India" and "Bengaluru" as geographical locations (GPE).

```
[47] spacy.explain("PERSON")
```

'People, including fictional'

```
displacy.render(text1,style="ent",jupyter=True)
```

Indian Space Research Organisation ORG (ISRO) is the space agency of India GPE. The organisation is involved in science engineering and technology to harvest the benefits of outer space for India GPE and the mankind. ISRO is a major constituent of the Department of Space ORG (DOS), Government of India ORG. The department executes the Indian Space Programme ORG primarily through various Centres ORG or units within ISRO ORG. ISRO was previously the Indian National Committee for Space Research (INCOSPAR ORG), set up by the Government of India ORG in 1962 DATE, as envisioned by Dr GPE Vikram Sarabhai GPE. ISRO was formed on August 15, 1969 DATE and superseded INCOSPAR with an expanded role to harness space technology. DOS ORG was set up and ISRO was brought under DOS ORG in 1972 DATE. The prime objective of ISRO/DOS is the development and application of space technology for various national needs. To fulfil this objective, ISRO ORG has established major space systems for communication, television broadcasting and meteorological services, resources monitoring and management, space-based navigation services. ISRO has developed satellite launch vehicles, PSLV ORG and GSLV ORG, to place the satellites in the required orbits. Alongside its technological advancement, ISRO contributes to science and science education in the country. Various dedicated research centres and autonomous institutions for remote sensing, astronomy and astrophysics, atmospheric sciences and space sciences in general function under the aegis of Department of Space ORG. ISRO's own Lunar GPE and interplanetary missions along with other scientific projects encourage promote science education, apart from providing valuable data to the scientific community which in turn enriches science. ISRO has its headquarters in Bengaluru GPE. Its activities are spread across various centres and units. Launch Vehicles PERSON are built at Vikram Sarabhai Space Centre FAC (VSSC), Thiruvananthapuram ORG. Satellites ORG are designed and developed at U R Rao Satellite Centre ORG (URSC), Bengaluru ORG. Integration ORG and launching of satellites and launch vehicles are carried out from Satish Dhawan Space Centre ORG (SDSC ORG), Sriharikota PERSON. Development of liquid stages including cryogenic CARDINAL stage is carried out at Liquid Propulsion Systems Centre ORG (LPSC), Valiamala & Bengaluru ORG. Sensors for Communication and Remote Sensing ORG satellites and application aspects of the space technology are taken

Figure 7.2: Output of Scanned Image NER

The Python code uses the spaCy library to perform named entity recognition (NER), which identifies and classifies names of people, organizations, places, and more in text. It loads a pre-trained English model and processes a text mentioning the Indian Space Research Organisation (ISRO). The code recognizes "ISRO," "Department of Space (DOS)," and "Vikram Sarabhai Space Centre (VSSC)" as organizations (ORG), and "Bengaluru" as a geographical location (GPE).

```
✓ 9s [1] import easyocr
reader = easyocr.Reader(['en'])
image_path = '/content/Name1.jpg' # Replace with your image file path
results = reader.readtext(image_path)
paragraph = ' '.join([result[1] for result in results])

WARNING:easyocr.easyocr:Neither CUDA nor MPS are available - defaulting to CPU. Note: This module is much faster with a GPU.

✓ 0s [29] print(paragraph)

MANSI

✓ 0s [30] import spacy
import spacy_transformers
from spacy import displacy

✓ 4s [31] NER = spacy.load("en_core_web_trf")

✓ 0s [32] text1= NER(paragraph)

✓ 0s [33] for word in text1.ents:
    print(word.text,word.label_)

MANSI PERSON

✓ 0s [34] displacy.render(text1,style="ent",jupyter=True)

MANSI PERSON
```

Figure 7.3: Output of Handwritten NER

The Python code uses the EasyOCR library to extract text from an image, a process called optical character recognition (OCR). It imports EasyOCR, creates a reader object, specifies the image file path, and uses the reader to extract and print the text. In this example, the code reads and prints the name "MANSI" from the image.


```
✓ [49] displacy.render(text1, style="ent", jupyter=True)
```

ISRO **ORG** was previously the Indian National Committee for Space Research **ORG** (INCOSPAR), set up
by the Government of India **ORG** in 1962 **DATE**, as envisioned by Dr. VikramA Sarabhai **PERSON** :
ISRO **ORG** has its headquarters in Bengalurud **GPE**

Figure 7.4: Output of Word text NER

The screenshot shows the website of the Indian Space Research Organisation (ISRO), India's government agency for space activities. The site provides details about ISRO's missions, activities, and structure. It mentions that ISRO was originally called the Indian National Committee for Space Research (INCOSPAR) and was founded by the Indian government in 1962, with its headquarters in Bengaluru.

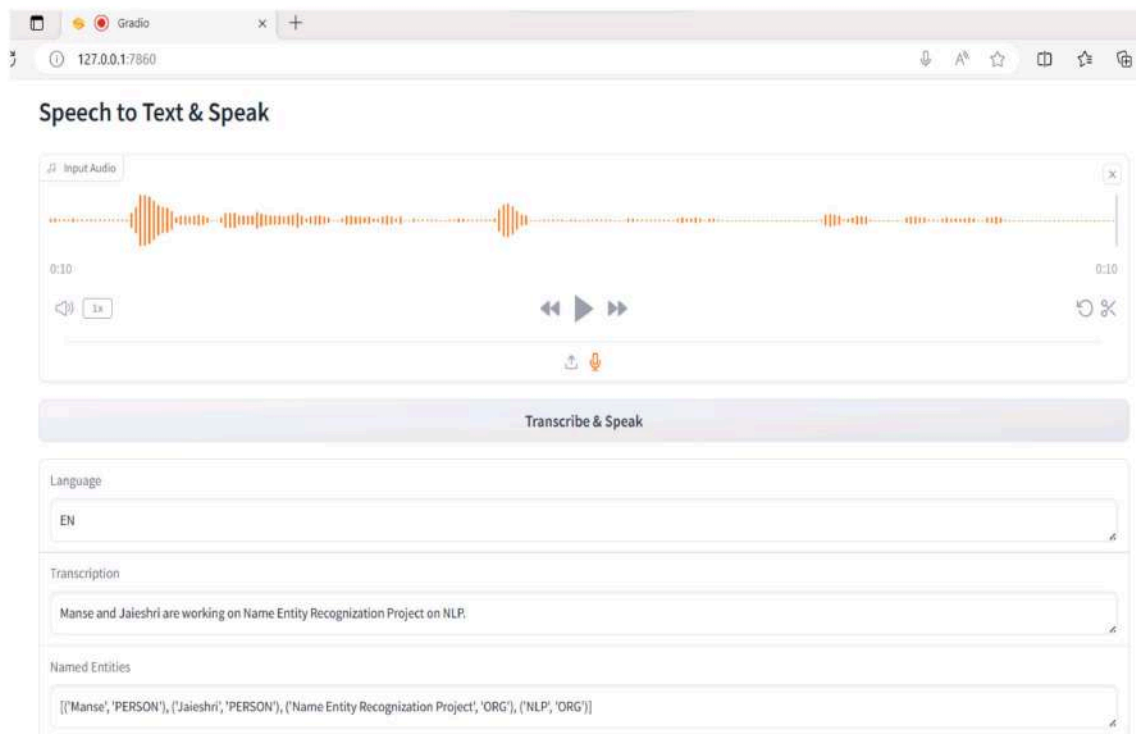


Figure 7.5: Output of Speech NER

The screenshot shows a website that can convert speech to text and text to speech. Users can speak into a microphone to see their words transcribed on the screen, or they can type text to hear it spoken aloud. In the screenshot, the text "Manse and Jaieshri are working on Name Entity Recognition Project on NIP" appears, indicating the website transcribed speech about a project on named entity recognition (NER). NER is a natural language processing task that identifies and classifies names of people, organizations, places, and more in text.

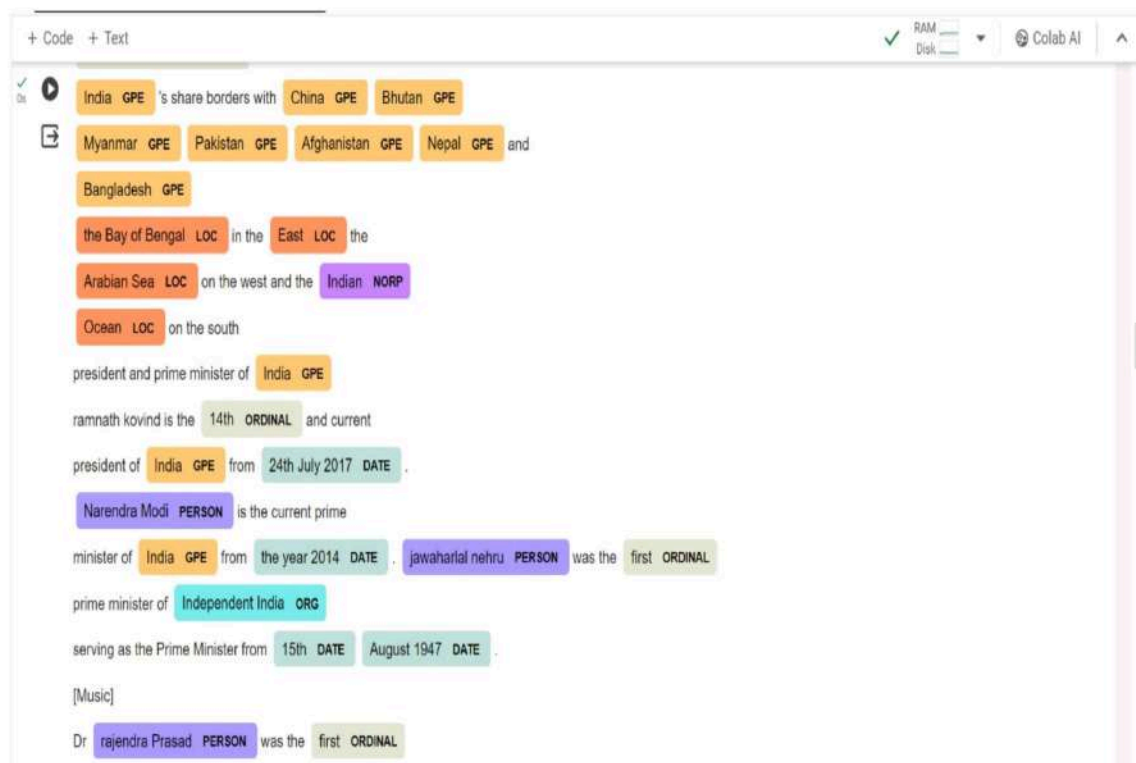


Figure 7.6: Output of Video NER

The screenshot shows a webpage with a code snippet related to geography, mentioning countries that border India, such as China, Myanmar, Pakistan, Bangladesh, and Bhutan, as well as bodies of water like the Bay of Bengal, the Arabian Sea, and the Indian Ocean. The code includes keywords like "LOC" for location and "ORDINAL" for order, which are commonly used in programming languages, though the specific language isn't clear.

```
✓ [15] doc = nlp("My credit card payment will be delayed")
2s   for ent in doc.ents:
      print(ent.text,ent.start_char,ent.end_char,ent.label_)
      displacy.render(nlp(doc.text),style='ent',jupyter=True)

credit card 3 14 PRODUCT
My credit card PRODUCT payment will be delayed

✓ [16] doc = nlp("What are the charges on credit card late payment in Bank of America")
0s   for ent in doc.ents:
      print(ent.text,ent.start_char,ent.end_char,ent.label_)
      displacy.render(nlp(doc.text),style='ent',jupyter=True)

credit card 24 35 PRODUCT
payment 41 48 ACTIVITY
America 60 67 ACTIVITY
What are the charges on credit card PRODUCT late payment ACTIVITY in Bank of America ACTIVITY

✓ [17] doc = nlp("I lost my investment account password and cannot open my account now")
0s   for ent in doc.ents:
      print(ent.text,ent.start_char,ent.end_char,ent.label_)
      displacy.render(nlp(doc.text),style='ent',jupyter=True)

investment account 10 28 PRODUCT
account 57 64 PRODUCT
I lost my investment account PRODUCT password and cannot open my account PRODUCT now
```

Figure 7.7: Output of Custom NER

The screenshot shows Python code using the spaCy library for named entity recognition (NER), a technique in natural language processing (NLP) to identify and classify names of entities in text. The code imports a pre-trained English model (en-core-web-sm) and applies it to sentences mentioning financial products like credit cards and investment accounts. The model recognizes these terms as "PRODUCT" entities and notes their positions in the sentences.

Chapter 8

Conclusion and Future Scope

8.1 Conclusion

This research paper presents, Named Entity Recognition (NER) as a pivotal advancement in information extraction, enabling the identification and extraction of crucial entities such as names, dates, and locations from a variety of media formats, including scanned images, audio recordings, and YouTube videos.

Moreover, the incorporation of NER techniques across multiple media types not only broadens the scope of information processing but also facilitates knowledge discovery. Our research has demonstrated the effectiveness of employing NER in tasks such as automated transcription and speech synthesis. This script handles audio processing, language detection, and named entity extraction, thereby serving as a versatile tool for automated transcription tasks.

Similarly, our script for extracting transcripts from Hindi YouTube videos underscores the importance of automation in fetching and analyzing content. By accurately extracting video IDs and leveraging external APIs for transcript retrieval, the script streamlines the process for researchers and analysts interested in studying Hindi content. Furthermore, the integration of NER techniques enhances the utility of the script by facilitating the identification and extraction of named entities from the transcripts, thereby enabling deeper analysis and insight generation.

In summary, our research offer valuable tools for researchers and analysts, facilitating tasks related to information extraction, transcription, and analysis across different media formats. The integration of NER techniques underscores the significance of our approach in enhancing data understanding and knowledge extraction from diverse sources. These findings contribute to the broader field of information extraction and data analysis, showcasing the potential of NER in advancing research and exploration across various domains.

8.2 Future Scope

In future scope, we're focusing on two main areas to improve Named Entity Recognition (NER). Firstly, we'll develop custom NER models tailored for specific domains or languages. These models will be fine-tuned to work better with domain-specific scanned documents, and we'll also explore creating new categories for better accuracy. Secondly, we aim to make NER work for multiple languages, including Hindi and Marathi, by converting text from images. Additionally, we're excited to explore NER's potential in extracting entities from handwritten texts. By addressing these areas, we hope to enhance NER's effectiveness across different languages and real-world scenarios.

In addition to the primary focus on improving Named Entity Recognition (NER) through domain-specific models and multilingual capabilities, our future scope involves delving deeper into the nuances of NER applications. One avenue we're exploring is the integration of advanced natural language processing techniques to enhance the contextual understanding of entities. By incorporating contextual information from surrounding text, we aim to refine entity recognition accuracy, especially in cases where entities may have multiple meanings or contexts.

Furthermore, we recognize the growing importance of entity linking in NER systems. Beyond merely identifying entities, linking them to relevant knowledge bases or ontologies can significantly enrich the understanding of text data. Our research will delve into techniques for seamless entity linking, ensuring that identified entities are accurately associated with pertinent information sources, thereby

enhancing the overall utility of NER outputs for downstream tasks such as information retrieval and knowledge graph construction.

Moreover, we're committed to addressing the challenges posed by noisy or ambiguous text data, which are prevalent in real-world applications of NER. Through robust preprocessing techniques and the integration of error-correction mechanisms, we aim to mitigate the impact of noise and ambiguity on NER performance. Additionally, we'll explore the utilization of semi-supervised and unsupervised learning approaches to adaptively learn from unannotated data, thereby enhancing the robustness of NER models across diverse linguistic contexts and data modalities.

Another area of interest lies in the exploration of multimodal NER, wherein information from different modalities such as text, images, and audio is leveraged synergistically to improve entity recognition accuracy. By incorporating visual cues from images or audio signals from spoken text, we aim to enhance the robustness and versatility of NER systems, particularly in scenarios where textual data alone may be insufficient or unavailable.

Overall, our holistic approach to advancing NER encompasses not only technical enhancements in model architectures and linguistic coverage but also considerations for real-world applicability, robustness, and adaptability across diverse data sources and modalities. Through continuous research and innovation, we strive to push the boundaries of NER capabilities and pave the way for more effective and scalable solutions in natural language understanding and information extraction.

Appendix A

Brief Biodata of each student

A.1 Student 1: Mansi Kumbhar

- **Name:** Mansi Kumbhar
- **Branch:** Data Science
- **Roll Number:** 2015038
- **Email:** kumbharmansi25@gmail.com
- **Mobile Number:** +91 7045155146
- **Skills:** Python, HTML, CSS



A.2 Student 2: Jayshree Parate

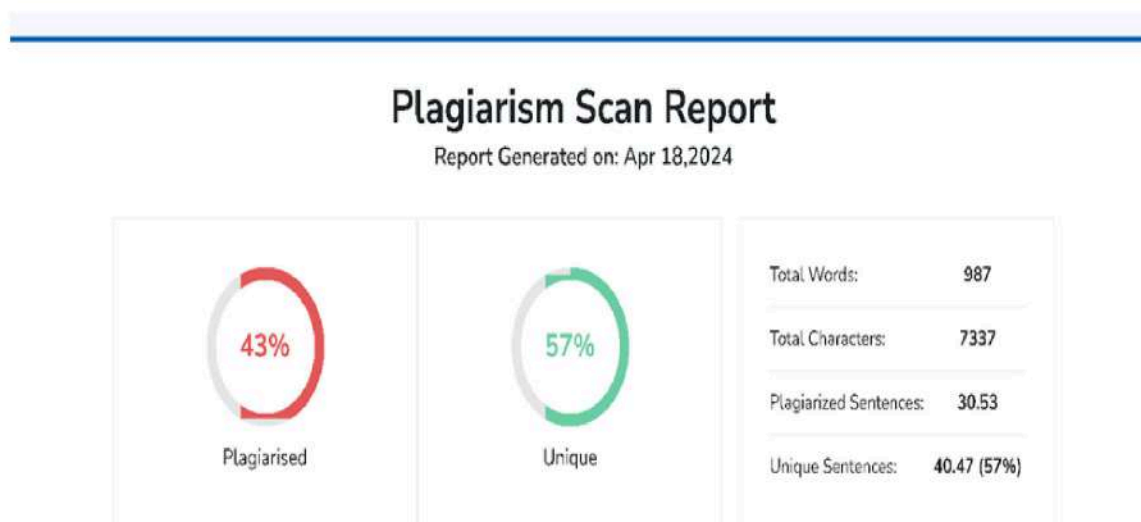
- **Name:** Jayshree Parate
- **Branch:** Data Science
- **Roll Number:** 2015050
- **Email:** jayshreeparate2002@gmail.com
- **Mobile Number:** +91 8591183305
- **Skills:** Python, HTML, CSS



Appendix B

Plagiarism Report

B.1 Plagiarism Report of Technical Paper



B.2 Plagiarism Report of Black Book



Plagiarism Checker X - Report

Originality Assessment

14%



Overall Similarity

Date: May 26, 2024
Matches: 1066 / 7536 words
Sources: 27

Remarks: Moderate similarity detected, consider enhancing the document if necessary.

Verify Report:
Scan this QR Code



NAMED ENTITY RECOGNITION FOR SCANNED IMAGES, AUDIO AND VIDEO

Prof.Aarti Dharmani
Department of Data Science
Usha Mittal Institute of Technology
Mumbai, Maharashtra
aarti170394@gmail.com

Mansi Kumbhar
Department of Data Science
Usha Mittal Institute of Technology
Mumbai, Maharashtra
kumbharmansi25@gmail.com

Jayshree Parate
Department of Data Science
Usha Mittal Institute of Technology
Mumbai, Maharashtra
jayshreeparate2002@gmail.com

Abstract—Identified as a critical function within natural language processing, Named Entity Recognition (NER) entails discerning and categorizing named entities from raw textual data. This project addresses the challenge of developing an effective NER system tailored specifically for scanned images. We propose a novel approach to accurately extract text from scanned images using Optical Character Recognition (OCR) techniques and subsequently apply a context-aware NER model to identify and categorize named entities. The system is designed to handle image quality issues, consider contextual information, and provide multilanguage support. We demonstrate the system's efficiency, scalability, and accuracy in extracting valuable named entity information from scanned images, opening new possibilities for applications in document analysis, information extraction, and archival processes.

Index Terms—Named Entity Recognition (NER), Scanned Images, Optical Character Recognition (OCR), Audio Processing, Video Processing, Information Extraction.

I. INTRODUCTION

Named Entity Recognition (NER) has emerged as a fundamental natural language processing (NLP) technique, initially developed to parse and categorize named entities within textual data. Traditionally, NER systems excelled at identifying entities such as persons, organizations, locations, and dates, aiding in various text-based applications like information extraction, question answering, and document summarization.

However, as the digital landscape evolves and the demand for extracting insights from diverse media formats intensifies, the limitations of conventional NER methodologies have become apparent. With the proliferation of digitized historical records, handwritten manuscripts, and printed materials, the necessity to extend NER capabilities beyond text has become evident. Scanned images, once solely the domain of visual processing techniques, now present a unique challenge for NER systems, requiring them to decipher handwritten text and detect printed characters to identify named entities accurately.

Moreover, the advent of multimedia content, including audio and video, has ushered in a new era for NER. In the realm of audio processing, NER must navigate the intricacies of speech

recognition, transcribing spoken content, and subsequently identifying relevant entities within the audio stream. Similarly, video data presents its own set of challenges, requiring NER systems to extract textual information from visual elements like subtitles, captions, or on-screen text, and associate them with specific entity types.

The integration of NER techniques with audio and video processing not only presents novel challenges but also opens up exciting opportunities. This fusion enables the extraction of valuable information from multimedia sources, facilitating applications such as multimedia content analysis, information retrieval, and archival processes. By extending NER beyond its traditional text-based confines to encompass audio and video, we aim to enhance the ability to identify and categorize named entities across a broader spectrum of media types, thereby unlocking new avenues for knowledge discovery and enrichment.

II. LITERATURE REVIEW

The text outlines various advancements and applications in the fields of computer vision, natural language processing (NLP), speech recognition, sentiment analysis, and music information retrieval (MIR). Computer vision involves the interpretation of visual data by computers, including tasks like handwriting recognition (HWR) using Python and OpenCV [2], and optical character recognition (OCR) for extracting machine-readable text from images [1]. NLP, a subset of artificial intelligence, focuses on machines understanding and generating natural language, with applications like video summarization to condense longer videos into concise summaries using techniques such as text extraction and audio analysis [3].

Additionally, the text discusses the study of polyglots and factors influencing their language acquisition abilities [4]. It highlights a case study of a polyglot English teacher proficient in nine languages, emphasizing the role of brain lateralization, learning strategies, age, and language contact in language acquisition. The subject's proficiency in languages, including the use of English as a bridging language, is examined along with

factors such as language aptitude, motivation, and cognitive factors contributing to successful language learning.

Furthermore, speech recognition technologies are explored, utilizing algorithms like recurrent neural networks (RNNs) and NLP sequence models to convert spoken words into text efficiently [5]. Speaker recognition involves identifying users based on voice characteristics, while sentiment analysis aims to identify emotions in text data using classification algorithms. These technologies have applications in various domains, including transcribing speeches, analyzing audio data, and identifying emotions expressed in text.

Lastly, the text introduces music information retrieval (MIR), a field at the intersection of musicology, digital signal processing, machine learning, and information retrieval [6]. It discusses the emergence of Python as a preferred language for MIR research, driven by factors like the availability of machine learning libraries and tools for dealing with text data. The development of *librosa*, a Python package for audio and music signal processing, aims to facilitate MIR research and make core techniques accessible to a broader community of scientists and programmers.

III. PROPOSED SYSTEM

Named Entity Recognition (NER) has long been a cornerstone of Natural Language Processing (NLP), primarily focused on identifying and categorizing entities like people, organizations, locations, and dates within textual data. However, with the evolving digital landscape, the need to extend NER beyond mere text has become evident. This expansion is essential to effectively parse through diverse media formats such as scanned images, handwritten manuscripts, audio, and video.

In response to these challenges, a proposed system aims to enhance NER capabilities by integrating it with audio and video processing techniques. For instance, in audio processing, the system would employ speech recognition algorithms to transcribe spoken content and subsequently identify relevant entities within the audio stream. Similarly, in video data analysis, the system would extract textual information from visual elements like subtitles or captions, associating them with specific entity types.

One of the key advantages of this proposed system is its ability to decipher handwritten text and detect printed characters within scanned images. This feature is particularly crucial for historical records and printed materials, where conventional NER methodologies may fall short.

In summary, our proposed system offers valuable tools for researchers and analysts engaged in information extraction, transcription, and analysis across different media formats. By integrating NER techniques, we highlight the significance of our approach in enhancing data understanding and knowledge extraction from diverse sources. These findings contribute to the broader field of information extraction and data analysis, demonstrating the potential of NER in advancing research and exploration across various domains.

IV. DESIGN AND IMPLEMENTATION DETAILS

A. Step 1

1) Image Input: Image input refers to the utilization of visual data, encompassing scanned documents, photographs, or images of handwritten text, as the primary source of information for subsequent processes. These images offer a visual representation of text or handwritten content, capturing nuances like font style, size, and layout. Their versatility allows them to be sourced from diverse mediums, including scanned documents, photographs, or digital images, making them adaptable inputs for processing systems.

However, challenges accompany image input. Variability in image quality due to factors such as resolution, lighting conditions, and distortion can significantly affect the accuracy of subsequent processing steps. Extracting text accurately from images, especially in cases of poor image quality or handwritten text, presents challenges for OCR systems. Additionally, recognition systems must contend with the diversity of languages, scripts, and writing styles present in images, necessitating robust language models and character recognition algorithms. Despite these challenges, image input remains a valuable resource for various applications, provided effective pre-processing techniques and recognition algorithms are employed.

2) Input Audio: Input audio, crucial for Automatic Speech Recognition (ASR) systems, comprises spoken language data essential for transcription. It encapsulates various characteristics and challenges. In terms of characteristics, input audio predominantly contains speech signals from one or multiple speakers, spanning conversations, lectures, phone calls, and more. Its variability presents challenges, ranging from diverse speaker accents, speech rates, background noise levels, to audio quality. ASR systems must exhibit robustness to handle such variations adeptly, ensuring accurate transcription. Additionally, input audio can be in multiple formats like WAV, MP3, AAC, necessitating ASR systems' adaptability to process diverse audio file types effectively.

However, challenges persist in the realm of input audio. Background noise, environmental sounds, and overlapping speech pose significant hurdles, diminishing speech recognition accuracy. Accents and dialects further complicate matters, as ASR systems may struggle with variations outside their training data's scope, potentially leading to inaccuracies. Moreover, rapid speech or irregular pacing present additional complexities for ASR systems to transcribe accurately.

3) Input Video: Input video, in the context of ASR (Automatic Speech Recognition), refers to the visual data accompanying audio that some systems can utilize to enhance transcription accuracy. This includes the video stream capturing facial expressions, lip movements, gestures, and other visual cues associated with speech. Video plays a crucial role in ASR by providing contextual information that aids in speech recognition. Lip movements, facial expressions, and gestures offer additional context, helping to disambiguate

speech, especially in noisy environments or when dealing with unclear audio. Moreover, visual cues can assist in speaker identification and tracking, which is particularly beneficial in scenarios involving multiple speakers or poor audio quality.

However, integrating video data with audio processing pipelines presents challenges, requiring additional computational resources and specialized algorithms. Privacy concerns also arise when processing video data, especially in situations where individuals may not consent to being recorded or monitored. Furthermore, while video can improve ASR accuracy, it introduces complexities such as variations in lighting conditions, camera angles, and occlusions. Thus, while input video holds promise for enhancing ASR performance, careful consideration of these challenges and considerations is necessary for effective implementation.

B. Step 2

1) Image preprocessing is an essential initial step in image analysis tasks, aimed at enhancing the quality of raw image data before it undergoes further analysis or processing by algorithms. One fundamental aspect is noise reduction, as images often contain unwanted noise that can distort crucial features. Additionally, image enhancement techniques are utilized to adjust contrast, brightness, and sharpness, thereby improving visual quality. Normalization of intensity values is also crucial to ensure data falls within a specific range, facilitating easier comparison or analysis of images. Resizing and cropping may be necessary for standardization, while color correction techniques such as white balancing or color space transformations ensure accurate representation of colors across different images.

Moreover, preprocessing involves feature extraction through methods like edge detection, corner detection, or blob detection, which extract relevant features for tasks like object recognition or image segmentation. By applying these preprocessing techniques, raw image data is refined and optimized, leading to enhanced accuracy and efficiency in subsequent image analysis algorithms.

2) The speech processing involves several steps. Initially, the system takes an audio file as input, expecting it to contain speech for transcription. Next, the input audio undergoes resampling to a standardized sample rate of 16 kHz using the librosa library. This resampling step ensures consistency in audio processing.

Subsequently, the resampled audio is processed using the Whisper model, a deep learning model specifically trained for speech recognition tasks. The Whisper model converts the resampled audio into a mel spectrogram, which is a visual representation of the audio spectrum. This mel spectrogram serves as the primary input for the speech recognition model. The Whisper model then transcribes the speech from the mel spectrogram into text format, representing the spoken content of the input audio. Following transcription, the system may perform additional processing steps such as language detection and named entity recognition, although these are not

explicitly specified in the request. Finally, the system outputs the transcribed text, providing the recognized speech content extracted from the input audio file.

C. Step 3

1) OCR (Optical Character Recognition) is a vital technology employed for extracting text from images. One such tool, Easy OCR, utilizes advanced algorithms to perform this task efficiently. Initially, the input image undergoes preprocessing to enhance its quality and improve recognition accuracy. Subsequently, the OCR engine employs pattern recognition techniques to analyze the image, identifying and recognizing individual characters, words, and sentences. This process involves comparing the visual patterns in the image against known character sets to decipher text accurately. Easy OCR can handle various fonts, languages, and writing styles, making them invaluable for digitizing printed documents, automating data entry tasks, and enabling text extraction from images in diverse applications such as document management, digital archives, and text analytics.

2) In the stage of speech recognition, specific characteristics of the speech audio are extracted to aid in distinguishing between different phonemes, which are the fundamental units of sound in a language. These extracted features, such as pitch and formants, play a crucial role in enabling the system to discern nuances in speech patterns. By analyzing these features, the speech recognition system can accurately interpret spoken words and phrases, facilitating effective communication between humans and machines. This process of feature extraction serves as a foundational step in the broader field of natural language processing, enabling advancements in various applications ranging from virtual assistants to automated transcription services.

D. Step 4

1) Extract Text (from OCR): Text extraction is a critical stage in the process of scanning images via Optical Character Recognition (OCR). When text is identified within the scanned image, this stage becomes essential for extracting that text data. By extracting the text, it becomes usable and accessible for various purposes such as indexing, searching, and analysis. This process enables the conversion of scanned documents into editable and searchable formats, facilitating easier information retrieval and manipulation. Additionally, text extraction enhances the efficiency of data processing and management by digitizing content that was previously only available in non-editable formats. Through OCR-based text extraction, organizations and individuals can unlock the potential of scanned documents, making them more versatile and valuable in the digital age.

2) Transcribe Audio: In the process of speech recognition, features extracted from the speech audio are pivotal. These features serve as the foundation for a speech recognition model to determine the most probable sequence of words corresponding to the audio input. By leveraging these extracted features,

3) **Extract Audio (from Video):** Extracting audio from a video is a crucial step in many multimedia tasks. When a video is inputted, the process involves separating the audio stream from the video stream. This separation allows for independent manipulation of the audio, such as editing, mixing, or applying effects. By extracting the audio, users can repurpose it for various purposes, such as creating podcasts, music tracks, or sound effects libraries. Additionally, it enables efficient storage and transmission of audio data, particularly in scenarios where only the audio content is needed, saving bandwidth and disk space. Overall, extracting audio from a video stream enhances flexibility and usability in multimedia production and distribution.

Named Entity Recognition (NER) - SpaCy: Named Entity Recognition (NER) is a crucial step in speech recognition systems like SpaCy, where it identifies and categorizes named entities within transcribed text. These entities encompass various elements such as people's names, locations, or organizations. Although not all automatic speech recognition (ASR) systems incorporate this step, NER plays a significant role in extracting pertinent information from the transcribed content, facilitating tasks like information retrieval and analysis. By accurately identifying and classifying named entities, NER enhances the overall understanding and usability of transcribed speech data.

The final output of the process is a list of extracted entities from the input image. These entities can include names of people, organizations, locations, dates, numbers, and other relevant information present in the text, audio, and video. This is the final output, which can include the transcribed text, any recognized named entities. Extracted entities provide crucial information that aids in various tasks such as information retrieval, data analysis, and content summarization. By identifying and organizing these entities, systems can efficiently process and interpret the underlying data, facilitating seamless integration into workflows and applications. This output serves as a valuable resource for understanding and utilizing the content contained within the input, contributing to enhanced automation and decision-making processes across different domains and industries.

```
graph TD; subgraph Row1 [ ]; direction LR; I1[Input Image] --> P1[Image Pre Processing]; I2[Input Audio] --> P2[Speech Recognition]; I3[Input Video] --> P3[Extract Audio]; end; subgraph Row2 [ ]; direction LR; P1 --> O[OCR - EasyOCR]; P2 --> T[Transcribe Audio]; P3 --> E[Extract Text]; end; subgraph Row3 [ ]; direction LR; O --> NER[NER - Spacy]; T --> NER; E --> NER; end; NER --> R[Extracted Entities];
```

The flowchart illustrates a three-stage NLP pipeline. The first stage, labeled 'Pre-Processing', takes 'Input Image', 'Input Audio', and 'Input Video' as inputs and outputs 'Image Pre Processing', 'Speech Recognition', and 'Extract Audio' respectively. The second stage, labeled 'Text Extraction', takes these pre-processed inputs and outputs 'OCR - EasyOCR', 'Transcribe Audio', and 'Extract Text'. The third stage, labeled 'NER', takes the outputs from the second stage and feeds them into 'NER - Spacy', which finally outputs 'Extracted Entities'.

[illegible][illegible]

In the above image, we input text in image format and check for entities.



Fig. 4. Output of Handwritten NER



Fig. 7. Output of Video NER

In the above image, we input handwritten text and check for entities.



Fig. 5. Output of Word text NER

In the provided image, we input text in various formats and check for entities.



Fig. 6. Output of Speech NER

In the above image, we input live audio recordings and check for entities.

In the above image, we input a YouTube video link and check for entities.



Fig. 8. Output of Custom NER

In the provided image, we created a customized category of Named Entity Recognition (NER).

V. CONCLUSION AND FUTURE SCOPE

This research paper presents, Named Entity Recognition (NER) as a pivotal advancement in information extraction, enabling the identification and extraction of crucial entities such as names, dates, and locations from a variety of media formats, including scanned images, audio recordings, and YouTube videos.

Moreover, the incorporation of NER techniques across multiple media types not only broadens the scope of information processing but also facilitates knowledge discovery. Our research has demonstrated the effectiveness of employing NER in tasks such as automated transcription and speech synthesis. This script handles audio processing, language detection, and named entity extraction, thereby serving as a versatile tool for automated transcription tasks.

Similarly, our script for extracting transcripts from Hindi YouTube videos underscores the importance of automation in fetching and analyzing content. By accurately extracting video IDs and leveraging external APIs for transcript retrieval, the script streamlines the process for researchers and analysts interested in studying Hindi content. Furthermore, the integration of NER techniques enhances the utility of the script by facilitating the identification and extraction of named entities from the transcripts, thereby enabling deeper analysis and insight generation.

In summary, our research offer valuable tools for researchers and analysts, facilitating tasks related to information extraction, transcription, and analysis across different media formats. The integration of NER techniques underscores the significance of our approach in enhancing data understanding and knowledge extraction from diverse sources. These findings contribute to the broader field of information extraction and data analysis, showcasing the potential of NER in advancing research and exploration across various domains.

In future scope, we're focusing on two main areas to improve Named Entity Recognition (NER). Firstly, we'll develop custom NER models tailored for specific domains or languages. These models will be fine-tuned to work better with domain-specific scanned documents, and we'll also explore creating new categories for better accuracy. Secondly, we aim to make NER work for multiple languages, including Hindi and Marathi, by converting text from images. Additionally, we're excited to explore NER's potential in extracting entities from handwritten texts. By addressing these areas, we hope to enhance NER's effectiveness across different languages and real-world scenarios.

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Mansi Kumbhar
Jayshree Parate